

Chapter 30 — Cutaneous Diseases of the External Genitalia

- Diagnosis and treatment of genital cutaneous disease is a **core element of urological practice**, yet is **often overlooked in formal urology residency training**.
- The subject lies at the **interface of several specialties**: urology, infectious diseases, rheumatology, allergy–immunology and dermatology.

INTRODUCTION TO BASIC DERMATOLOGY

Diagnostic philosophy

- Diagnosis rests critically on **history and physical examination**; laboratory testing is **peripheral and confirmatory**.
- Visual inspection alone often **narrows the differential substantially**.
- The skin has a **limited repertoire of morphological expression**, so many diseases mimic one another. The urologist should therefore have a **low threshold for skin biopsy or targeted laboratory tests** to separate clinical mimics.

SKIN LAYERS (FIG. 30.1)

Layer	Composition	Key points
Epidermis	Stratified squamous epithelium	Thickness 0.05–1.5 mm depending on site; melanocytes populate the lower layers
Dermis	Collagen, elastin, reticular fibres	Papillary dermis (thin, superficial) over reticular dermis (thick, deep); contains blood vessels and nerves
Hypodermis (subcutaneous tissue)	Largely fat	Deepest layer

History

- Hundreds of dermatoses can involve the genitalia, and each evolves in appearance over time, so a **methodical, systematic approach** is essential.
- The history covers **duration, rate of onset, location, associated symptoms, family history, allergies, occupation and previous treatment**.
- Common symptoms are **pruritus (itching), burning, stinging and pain**.
- The **absence of a symptom** (for example, a painless lesion) is diagnostically informative and should be recorded.

Examination

- Examine the **distribution of primary and secondary lesions** with a **full skin survey**, not the genital skin alone.
 - Example: red, silvery-scaled genital plaques plus similar plaques on the **extensor surfaces** point to **psoriasis**.
- The **primary lesion is the key to diagnosis**. Describe its **colour** (red, brown, black, yellow, white, blue, green) and its **morphology**.
- Because genital skin is mucosal in character, **papules and macules may present as erosions**.
- **Secondary lesions** arise as the disease evolves, or from **scratching, rubbing or superinfection**.

TABLE 30.1 PRIMARY CUTANEOUS LESIONS (HABIF, 2004)

Category	Lesion	Defining feature	Size cut-off
Flat	Macule	Circumscribed flat discolouration (brown, blue, red or hypopigmented)	—
Elevated, solid	Papule	Elevated solid lesion of variable colour; may coalesce into plaques	Up to 0.5 cm
	Nodule	Circumscribed elevated solid lesion	>0.5 cm
	Plaque	Circumscribed elevated superficial solid lesion	>0.5 cm
Fluid-filled	Vesicle	Circumscribed collection of free fluid	≤0.5 cm
	Bulla	Circumscribed collection of free fluid	>0.5 cm
	Pustule	Collection of leucocytes and free fluid (pus)	—
	Wheal (hive)	Firm erythematous plaque from dermal infiltration with fluid ; may be transient	—

- **What the table tests:** a **single 0.5 cm threshold** divides papule from nodule or plaque, and vesicle from bulla.
- **Nodule versus plaque:** both are solid and larger than 0.5 cm; the **plaque is defined as superficial**.
- **Trap:** the **wheal is fluid-filled but is not a free-fluid collection** — it is oedematous dermis, and may vanish.

TABLE 30.2 SECONDARY CUTANEOUS LESIONS (HABIF, 2004)

Lesion	Definition	Depth / outcome
Scale	Excess dead epidermal cells from abnormal keratinisation and shedding	Surface
Crust	Dried serum and cellular debris (scab)	Surface
Erosion	Focal loss of epidermis	Does not breach the dermoepidermal junction → heals without scarring
Ulcer	Focal loss of epidermis and dermis	Heals with scarring
Fissure	Linear loss of epidermis and dermis with sharply defined vertical walls	Dermal
Atrophy	Depression from thinning of epidermis or dermis	—
Scar	Abnormal connective tissue formation	Implies dermal damage

- **What the table tests:** **depth decides scarring**. The **dermoepidermal junction** is the dividing line; anything reaching the dermis (ulcer, fissure, scar) implies permanence.
- **Paired contrast:** **erosion (epidermal, no scar)** versus **ulcer (dermal, scars)**.

BEDSIDE LABORATORY TESTS

Test	Target	Principle / note
Potassium hydroxide (KOH) preparation	Dermatophytes, <i>Candida</i> spp.	Dissolves keratin , leaving fungal hyphal walls prominent
Periodic acid–Schiff (PAS) stain	Dermatophytes, <i>Candida</i> spp.	Applied to scraped or touch specimens
Mineral oil preparation	Mites and eggs	Direct visualisation
Tzanck preparation	Herpes simplex, varicella zoster, molluscum contagiosum	Cytology of viral lesions

Biopsy

- **Indications:** difficult cases, or **suspected malignancy**.

- **Techniques:** curettage, punch, shave, incisional, complete excisional.
- **Setting:**
 - **Small scrotal or penile shaft lesions** → office, under **local anaesthesia**.
 - **Larger lesions or those involving the urethral meatus** → **operating theatre**.
- A 2–3 mm punch is often diagnostic; the defect is closed with **one or two 6-0 or 7-0 nylon sutures**, or left to **heal by secondary intention**.

ADJUNCTIVE DIAGNOSTIC TESTS

Test	Example use
Serology	Syphilis
Culture	<i>Pseudomonas aeruginosa</i>
Wood's ultraviolet lamp	Erythrasma (fluorescence); vitiligo (accentuated depigmentation)
Immunohistochemistry	Cytokeratin subtypes associated with lichen sclerosus variants

Concept — *Because the skin has a **limited morphological repertoire**, **precise description of the primary lesion** narrows the differential, and a **targeted bedside test** or **small punch biopsy** then separates the mimics.*

DERMATOLOGIC THERAPY

- Medical therapy spans a **broad range of topical and systemic compounds**.

SYSTEMIC AGENTS

Class	Agents named	Role / caveat
Antibiotics	—	Overprescribed by urologists unfamiliar with dermatoses
Antifungals	Fluconazole, ketoconazole, terbinafine	Restricted indications (see below)
Antivirals	—	Disease-specific
Anti-inflammatories	Glucocorticosteroids (GCS): prednisone, methylprednisolone	Short courses preferred
Antipruritics	Hydroxyzine, diphenhydramine (oral antihistamines); doxepin (tricyclic); selective serotonin reuptake inhibitors (SSRIs)	Symptomatic
Chemotherapeutic and biologic agents	Methotrexate, cyclophosphamide, adalimumab, etanercept, infliximab, ustekinumab	Covered under specific diseases
Immunosuppressants	Azathioprine, cyclosporine, tacrolimus	Covered under specific diseases
Other	Hydroxyurea	Covered under specific diseases
Disease-modifying antirheumatic drugs (DMARDs): Janus kinase (JAK) inhibitors	Baricitinib, tofacitinib, upadacitinib, ruxolitinib	Most recent class

Systemic antibiotics — caution

- Unfamiliarity with genital dermatoses **lowers the threshold for systemic antibiotics**.
- Their risks exceed those of topical agents:
 - **Resistant organisms**
 - **Drug interactions**

- **Disruption of bowel and vaginal flora**

- Altered flora and susceptibility patterns can **persist for protracted periods**, which demands **truly appropriate antibiotic use**.

Systemic antifungals

- The same caveats apply. **Superficial dermatophytosis (tinea cruris)** responds to **diligent topical antifungal therapy**.
- **Systemic antifungals are indicated only for:**
 - **Very extensive** cutaneous dermatophytosis
 - **Endemic mycoses** with skin involvement
 - **Majocchi granuloma** (deep follicular infection)
 - **Severe immunocompromise**
 - Infection **resistant to topical therapy**, even in immunocompetent patients
- **Ketoconazole:** avoid routine systemic use for superficial infection because of unpredictable **life-threatening hepatotoxicity and adrenal insufficiency** (United States Food and Drug Administration [FDA], 2013).

Systemic glucocorticosteroids

- **Pharmacokinetics:**
 - Absorbed in the **jejunum**; peak plasma level at **30–90 minutes**.
 - Plasma half-life is **1–5 hours**, but the **duration of effect is 8–48 hours**, depending on the agent.
- **Mechanism:**
 - They **release neutrophils from marrow but inhibit their migration** to inflamed tissue.
 - They **impair T-cell activation and antigen presentation by dendritic cells**.
- **Short-term use (≤ 3 weeks)**, for example allergic contact dermatitis: give a **single morning dose** to minimise **hypothalamic–pituitary–adrenal (HPA) axis suppression**.
- **Prednisone is the GCS of choice** (low cost, intermediate duration of action, variety of dosage forms). **Methylprednisolone** may be substituted to **reduce mineralocorticoid effects**.
- **Long-term adverse effects:** osteoporosis, cataract, hypertension, obesity, hyperglycaemia, **avascular necrosis of the femoral head**, immunosuppression, psychiatric change.
- Therefore **prefer topical steroids whenever clinically feasible**.

Topical therapy

- Topical agents are the **mainstay** for genital dermatoses; urologists are generally less familiar with them than dermatologists.
- **Five classes: emollients, anti-inflammatories, antibiotics, antifungals, chemotherapeutic agents.**
- The **vehicle determines the rate of absorption** of the active ingredient.

Emollients

- They **restore water and lipids** to the epidermis and are used in **dry-skin disease**.
- **Apply to moist skin** (after bathing) for maximal effect.
- **Ointment versus cream: ointments are preferred** because **creams more often irritate or burn**.
- **Urea and lactic acid** preparations are **particularly potent hydrating agents**.
- **Ceramides** (fatty acid plus sphingoid base) are the **main intercellular lipids of the outermost skin layer**. They are critical for hydration and **barrier function**, so ceramide formulations suit **xerosis (dryness)**.

Topical corticosteroids

- They are **potent anti-inflammatory agents** available in many preparations and strengths.
- **Adverse effects** arise from both **systemic absorption** and **local application**:
 - **Epidermal atrophy**
 - **Striae of the upper inner thigh**
 - Dermal change (**telangiectasia, hypopigmentation**)
 - **Allergic reactions**
 - **Altered course of skin infections and infestations**
- **Male genital skin is at higher risk of steroid atrophy** because of its **thin dermis**.
 - Atrophy is **usually reversible over several months** but **may be permanent**.
- **Under the foreskin**, the prepuce acts as an **occlusive dressing** and **enhances penetration**, so atrophy is especially likely (Fig. 30.2).

Topical JAK inhibitors

- **Ruxolitinib** is the example. It **blocks the pro-inflammatory JAK signalling pathway** without the **local adverse effects of long-term corticosteroids**.

PHYSICAL MODALITIES

Modality	Indications	Key points
Broadband and narrow-band ultraviolet B (UVB)	Atopic dermatitis (AD), psoriasis, seborrhoeic dermatitis (SD), vitiligo	Broadband full-body boxes carry a non-melanoma skin cancer risk
Single-wavelength UVB laser (308 nm)	Localised genital psoriasis or vitiligo	Small spot size; believed not to carry the non-melanoma skin cancer risk
Psoralen plus ultraviolet A (PUVA)	Psoriasis, vitiligo, AD, lichen planus (LP)	Phototoxic effect; substantial squamous cell carcinoma (SCC) risk with prolonged use → supplanted by narrow-band UVB boxes and lasers
Photodynamic therapy	Inflammatory, malignant and infectious dermatoses; large or resistant condyloma acuminata; genital warts in pregnancy	Cytotoxic oxygen radicals from photoactivated molecules; used alone or with cryosurgery, carbon dioxide (CO ₂) laser ablation or curettage; no established optimum regimen for off-label use
Lasers (CO₂, pulsed dye)	Condyloma acuminata; pearly penile papules	Smaller role overall
Cryosurgery	Genital and suprapubic molluscum contagiosum	More widely used than laser

- The main **drawback of narrow-band UVB and laser therapy** is the **inconvenience of weekly or twice-weekly sessions**.

Figure notes

- **Fig. 30.2 Steroid atrophy of penile shaft skin after 8 weeks of corticosteroid under the foreskin** (Habif, 2004).

Concept — *Therapy escalates from topical to systemic only when necessary. Every systemic agent carries risks that topical agents do not, while the **thin, occluded genital dermis** makes even topical steroid a double-edged tool.*

ALLERGIC DERMATITIS

- **Allergic contact or eczematous dermatitis** is a group of allergy-mediated processes producing **pruritic lesions**.

BOX 30.1 DIFFERENTIAL DIAGNOSIS OF ALLERGIC DERMATITIS (MARGOLIS, 2002)

- **Ecematous and dermatitic:** eczema, allergic dermatitis, contact dermatitis, irritant dermatitis, seborrhoeic dermatitis, intertrigo.
- **Genital-specific inflammatory:** balanoposthitis, Zoon balanitis.
- **Infective:** candidal illness, **impetigo**, **herpes simplex**, **herpes zoster**.
- **Iatrogenic:** drug reaction.
- **What the box is for:** a pruritic red genital eruption is **not automatically allergic** — **infection (candidal, herpetic, impetigo)** and **drug reaction** must be excluded first.

Atopic Dermatitis (Eczema)

Definition, lesion and epidemiology

- AD is a **chronic relapsing dermatitis** with a **predilection for flexures**, **intense pruritus** and **epidermal damage**.
- **Lesion:** **erythematous papules and thin plaques with secondary excoriations** (Fig. 30.3).
- **Border contrast:** the AD border is **ill-defined**, whereas **papulosquamous disorders** are **sharply bordered**.
- **Onset:** any age, but **90% present before age 5 years**.
- There is **susceptibility to irritants** (fragrances, preservatives, proteins) and a **propensity to asthma and allergic rhinitis**.

Genetics

- **Strong familial clustering** of atopy; **monozygotic twin concordance far exceeds dizygotic**.
- There is **no single marker gene**; **at least 11 loci** are associated.
- The key defect is the **inability to synthesise functional filaggrin**, producing a **"leaky" epithelial barrier** plus **chronic immune activation**.

Pruritus and superinfection

- **Intense pruritus is the hallmark**, and **controlling the urge to scratch is critical** to successful treatment.
- Itch is **worse in the evening**, aggravated by **sweating, occlusive undergarments and wool**.
- Scratching leads to ***Staphylococcus aureus* superinfection**; **bacterial toxins act as superantigens** that sustain the inflammatory cascade.

Diagnosis

- There is **no pathognomonic test, biopsy result or single clinical feature**.
- A biopsy showing **spongiotic dermatitis** is **suggestive**.
- A **personal or family history of atopy** is the **critical clue**.
- With genital AD, **extragenital involvement is commonplace**.

Triggers

- Chemicals, detergents and **house dust mites**; remove them on an **individualised basis**.

TREATMENT OF ATOPIC DERMATITIS

Agent / measure	Class / target	Role
Non-alkali soaps or soap substitutes	Cleansing	Gentle cleaning
Emollients	Barrier repair	Frequent use; avoid frequent hot bathing (evaporation triggers AD)
Soaks	—	Only during bacterial superinfection; stop once resolved
Topical corticosteroids	Anti-inflammatory	Short courses with rapid taper (risk of atrophy, dyschromia)
Tacrolimus, pimecrolimus	Topical calcineurin inhibitors	Effective; steroid-sparing in long-term therapy
Ruxolitinib	Topical JAK inhibitor	Adjunct
Diphenhydramine; cetirizine, loratadine	Sedating versus non-sedating antihistamines	Break the itch-scratch cycle, especially at bedtime
Oral antistaphylococcal agents	—	No significant benefit in a randomised double-blind trial
Azathioprine, systemic corticosteroids, cyclosporine, methotrexate, mycophenolate mofetil	Systemic immunosuppression	Rarely , for severe widely disseminated disease
Dupilumab	Monoclonal antibody, T helper 2 (Th2) pathway	First FDA-approved monoclonal antibody for AD; moderate-to-severe disease
Tralokinumab	Human monoclonal antibody against interleukin-13 (IL-13)	Moderate-to-severe AD

Concept — AD is fundamentally a **barrier disease**. The **flaggrin defect** leaks the barrier, which invites irritants and immune activation; itch then drives scratching, which breaches the barrier further and admits staphylococcal amplification — so every treatment either **repairs the barrier** or **breaks the itch-scratch cycle**.

Contact Dermatitis*Shared features*

- There are **two distinct entities**: **irritant contact dermatitis (ICD)** and **allergic contact dermatitis (ACD)**.
- Their **mechanisms differ** but their **presentations are similar**; the affected area is usually **sharply limited to the zone of exposure**.
- **Primary treatment**: **identify and reduce exposure** to the offending agent.

IRRITANT VERSUS ALLERGIC CONTACT DERMATITIS

Feature	Irritant contact dermatitis	Allergic contact dermatitis
Mechanism	Direct cytotoxic effect of an irritant chemical	Type IV (delayed) hypersensitivity in a previously exposed and sensitised individual
Share of contact dermatitis	~80%	Minority; genital involvement rare (2.4%)
Typical agents	Soaps, solvents, metal salts, acid- or alkali-containing compounds	Nickel sulfate (commonest, NACDG) ; textile dyes; topical antibiotics (neomycin, bacitracin); perfumes and fragrances; formaldehyde-releasing preservatives ; condom latex ; topical corticosteroids
Determinants	Substance identity, duration, concentration, temperature, pH , site	Prior sensitisation

Feature	Irritant contact dermatitis	Allergic contact dermatitis
Acute picture	Peaks within minutes to hours , then heals; burning, stinging, soreness; erythema, oedema, bullae or necrosis in a sharply defined area	Well-demarcated pruritic eruption; blistering or weeping
Chronic picture	Repeated subthreshold insults; pruritus commoner; less well demarcated	Scaly plaques
Confirmation	Exposure history	Patch testing (eFig. 30.4)
Treatment	Avoidance : protective clothing, safe occupational practice, barrier preparations (ointments, emollient creams, protective foams)	Allergen removal plus oral antihistamines; severe ACD → 3-week prednisone taper, not a short course
Wider significance	Occupational ICD is a major public-health cost (about \$1 billion per year in the United States)	Common reason for outpatient dermatology visits

Barrier products and allergen history

- Commercial barrier products named: **Atopiclair, Biafine, EpiCeram, MimyX, Neosalus Foam, PruMyx.**
- **North American Contact Dermatitis Group (NACDG) allergen lists:**
 - Derived from patch-test results; successive lists contain **largely the same allergens.**
 - **Patch testing sequence:** apply potential allergens at **known concentrations** in a **grid template** → read the reaction → confirm the diagnosis and identify the allergen. It is **generally performed by dermatologists.**
- **Nickel:**
 - Found in **costume jewellery and belt buckles** (eFig. 30.5).
 - Classically causes **earlobe dermatitis** from pierced earrings.
 - It is now a potential cause of **genital ACD** because of the **rising prevalence of genital piercing.**
- **The urologist must always ask about over-the-counter products:** genital moisturisers, **spermicides**, anti-yeast and anti-itch preparations, and **sexual lubricants.**

Concept — Contact dermatitis **maps to the footprint of exposure**. ICD is **dose-driven chemical injury** that can occur in anyone, whereas **ACD is an immunological reaction** only in the previously sensitised; both are treated first by **removing the agent**.

Erythema Multiforme and Stevens–Johnson Syndrome

Classification

- **Erythema multiforme (EM)** is a **generalised eruption** that may involve the genitalia, divided into **minor and major forms.**
- **EM major** was **formerly called Stevens–Johnson syndrome (SJS)**; it remains **controversial** whether they are **distinct entities or a spectrum.**

EM MINOR VERSUS STEVENS–JOHNSON SYNDROME

Feature	EM minor	SJS (EM major)
First description	Ferdinand von Hebra, 1860 (Austrian dermatologist)	—
Principal trigger	Herpes simplex virus types 1 and 2 (HSV-1, HSV-2) in most recurrent cases; herpetic lesions precede target lesions by 10–14 days	Drugs (commonest); rarely infectious

Feature	EM minor	SJS (EM major)
Implicated drugs	—	Non-steroidal anti-inflammatory drugs (NSAIDs); sulfonamides (especially co-trimoxazole); tetracycline, doxycycline; penicillins, cephalosporins; anticonvulsants
Prodrome	—	Upper respiratory tract illness (fever, cough, rhinitis, sore throat, headache), progressing abruptly after 1–14 days
Lesions	Abrupt, symmetrical, fixed red papules evolving into target lesions; grouped , anywhere including genitalia; predilection for oral mucosa, palms and soles	Red macules → blisters → epidermal necrosis; burn-like ; the severe form mimics toxic epidermal necrolysis (TEN)
Genital involvement	May occur (Fig. 30.6A)	About 60%: erythema and erosions of labia, penis and perianal region (Fig. 30.7)
Diagnosis	Clinical, not histological	Clinical
Course	Self-limited ; spontaneous resolution over several weeks without sequelae ; recurrence common , longer and more frequent in the immunosuppressed	Protracted, 4–6 weeks; mortality approaching 30% at the TEN-overlap end of the spectrum
Sequelae	None	Scarring: joint contractures, labial synechiae, vaginal stenosis, urethral meatal stenosis, anal strictures
Treatment	Oral antihistamines for symptoms; continuous suppressive acyclovir may prevent episodes , but acyclovir given after target lesions appear is of no benefit	See sequence below

- **SJS management sequence: immediate withdrawal of the offending drug → supportive care as for severe burns → admission to an intensive care or burn unit (reduces morbidity and mortality) → multispecialty team with burn-centre wound care.**
- There is **no strong evidence for any specific drug therapy**, and **systemic corticosteroids are controversial**.
- Options with limited randomised data:
 - **Cyclosporine 3–5 mg/kg/day**
 - **Tumour necrosis factor-alpha (TNF-α) inhibitors** such as **etanercept (Enbrel)**
 - **Plasmapheresis**
 - **Intravenous immunoglobulin (IVIG)**, especially noted

Figure notes

- **Fig. 30.6 EM: (A) targetoid lesions of the hands and penis** (Korting, 1981); (B) histology — **normal stratum corneum, necrotic epidermal keratinocytes, lymphoid infiltrate** (Elston and Ferring, 2009).

Concept — EM minor is infection-driven (HSV), self-limiting and clinically diagnosed, whereas SJS is drug-driven, burn-like mucocutaneous necrosis — so the first is prevented by viral suppression, and the second is treated by drug withdrawal and burn care.

PAPULOSQUAMOUS DISORDERS

- A disparate group of diseases sharing one **primary lesion: scaly papules and plaques**.

BOX 30.2 DIFFERENTIAL DIAGNOSIS OF PAPULOSQUAMOUS LESIONS (MARGOLIS, 2002)

- **Inflammatory:** psoriasis, seborrhoeic dermatitis, **pityriasis rosea**, **discoid lupus**, lichen planus, **fixed drug eruption**, reactive arthritis.
- **Infective:** dermatophyte infection, erythrasma, secondary syphilis, pityriasis versicolor.
- **Neoplastic or premalignant:** mycosis fungoides, Bowen disease, extramammary Paget disease.
- **What the box is for:** a scaly genital plaque that **fails to respond to "psoriasis" treatment** must prompt consideration of **infection (KOH preparation, syphilis serology)** and **malignancy (biopsy)**.

Psoriasis*Epidemiology and pathogenesis*

- Affects **up to 2% of the population**. Predisposition is **likely polygenic**.
- **Triggers** of a flare:
 - **Trauma**
 - **Streptococcal infection**
 - **Psychological stress**
 - **New medications**
- **One-third** of patients have a **family history**.

Lesion and distribution

- **Characteristic lesion:** a sharply demarcated erythematous plaque with **silvery-white scale**, produced by **rapid epidermal proliferation**.
- **Distribution** ranges from **elbows and knees alone** to the **entire skin surface**.
- **Bimodal onset:** **20–30 years** and **50–60 years**.
- **Quality-of-life burden** comes from pruritus, bleeding, and the cosmetic and psychosocial impact.

Genital psoriasis

- **Relatively common**, but usually within **generalised disease**.
- Patients may present **fearing malignancy or a sexually transmitted disease (STD)**.
- It **impairs self-esteem and sexual self-image**, particularly in **females**.
- **Diagnostic clues elsewhere:** elbows, knees, buttocks, **nails, scalp, umbilicus** (Fig. 30.8).
- **Inverse psoriasis:** in the **inguinal folds and intergluteal cleft**, **scale may be absent**. For a non-scaling fold plaque, **exclude tinea or Candida** by KOH preparation or fungal culture.
- **Circumcised versus uncircumcised:** plaques sit on the **glans and corona** if circumcised, and are **hidden under the prepuce** if uncircumcised.
- It may involve the **entire penis and scrotum** (Fig. 30.9).
- **Histology** — "**sandwich sign**" (Fig. 30.8B): **alternating neutrophils and parakeratosis in the stratum corneum**.

Treatment of genital psoriasis

- The disease is **chronic, relapsing and remitting**; a **substantial proportion of patients are frustrated** by the ineffectiveness of current treatment.

TREATMENT LADDER FOR GENITAL PSORIASIS

Step	Agents	Notes
First-line	Low-potency topical corticosteroid creams, short courses: e.g. 3% liquor carbonis detergens (tar) in 1% hydrocortisone cream; hydrocortisone butyrate 0.1%	No more than 2 weeks continuously on thin genital skin or occluded folds
Thick plaque	Triamcinolone 0.1% initially	Stronger steroid
Other topicals	Vitamin D ₃ analogues (calcitriol, calcipotriene); topical calcineurin inhibitors (pimecrolimus cream, tacrolimus ointment); low-potency retinoids	Sometimes too irritating or insufficiently effective
Phototherapy	308-nm excimer laser (approved for psoriasis)	Localised disease
PUVA	Oral psoralen plus UVA	Dose-dependent genital SCC risk after high-dose PUVA for body psoriasis → genital shielding strongly recommended → contraindicated for genital-localised psoriasis
Systemic (extensive disease)	See table below	—
Experimental	Vitamin D receptor ligands; antibodies or antisense oligonucleotides against T-lymphocyte surface molecules, TNF or intercellular adhesion molecules	Promising

SYSTEMIC AGENTS FOR EXTENSIVE PSORIASIS

Agent	Target / class
Methotrexate	Antimetabolite
Cyclosporine	Calcineurin inhibitor
Retinoids	Retinoid
Apremilast	Phosphodiesterase 4 (PDE4) inhibitor
Adalimumab, etanercept, infliximab	TNF- α inhibitors
Ixekizumab	Interleukin-17 (IL-17) inhibitor — best data for genital psoriasis
Ustekinumab	IL-12/23 inhibitor

- **Pattern across the table:** biologics are grouped by cytokine target; ixekizumab (IL-17) is the agent singled out for genital disease.

Concept — Genital skin *inverts the usual psoriasis rules*: scale is lost in folds (so **fungal infection must be excluded**), thin skin limits steroid strength and duration, and **PUVA is barred** because of SCC — so genital treatment is **low-potency, short, and non-carcinogenic**.

Reactive Arthritis (Formerly Reiter Syndrome)*Components and epidemiology*

- The syndrome comprises **urethritis, arthritis, ocular findings, oral ulcers and skin lesions**; only about **one-third** show all features.
- It is **commoner in males**, rarely diagnosed in children, and **commoner in human immunodeficiency virus (HIV)-positive patients**.

Triggers

- **Urethritis:** *Chlamydia*, gonococcus, *Ureaplasma*.
- **Gastrointestinal infection:** *Yersinia*, *Salmonella*, *Shigella*, *Campylobacter*.
- **Strong association with human leukocyte antigen (HLA)-B27;** whether **molecular cross-reactivity** between bacterial antigens and HLA-B27 causes autoimmunity **remains controversial**.

Extracutaneous features

- **Ocular:** conjunctivitis is commonest; iritis, uveitis, glaucoma and keratitis also occur.
- **Musculoskeletal:** polyarthritis and sacroiliitis are commonest; **chronic disability in a small minority**.

Skin features

- **Balanitis circinata:** scaly erythematous psoriasiform lesions of the penis (Fig. 30.11).
- **Keratoderma blennorrhagicum:** similar lesions on the soles.
- These are **difficult to distinguish from psoriasis;** histology cannot consistently separate them and may show psoriasiform dermatitis.
- **Differential diagnosis:** pustular psoriasis, gonococcal infection, Behçet syndrome.

Course and treatment

- **Genital disease is usually self-limited,** lasting weeks to months; it responds to **low-potency topical corticosteroids,** and **systemic therapy is rarely required**.
- **Sole lesions are more persistent** and respond to **potent topical retinoid (tazarotene)** or **systemic retinoid (acitretin)**.

GENITAL PSORIASIS VERSUS BALANITIS CIRCINATA

Feature	Genital psoriasis	Balanitis circinata (reactive arthritis)
Context	Generalised psoriasis; family history	Preceding urethritis or enteric infection; HLA-B27
Morphology	Sharply demarcated silvery-scaled plaque	Psoriasiform; coalescent lesions forming a wavy pattern (Fig. 30.10), though this may be absent (Fig. 30.11)
Histology	Sandwich sign	Psoriasiform dermatitis — cannot reliably distinguish
Associated findings	Nails, scalp, umbilicus	Conjunctivitis, arthritis, sacroiliitis, keratoderma blennorrhagicum
Course	Chronic relapsing	Self-limited over weeks to months
Treatment	Low-potency short-course steroid → systemic or biologic	Low-potency steroid; systemic rarely

Concept — *Balanitis circinata is psoriasis-like skin with a different history: the diagnosis is made from context (infection, HLA-B27, eye and joint), because morphology and histology may be indistinguishable.*

Lichen Planus

Definition and pathogenesis

- LP is the **prototype lichenoid dermatosis:** an **idiopathic inflammatory disease** of skin and mucous membranes.
- **Lichenoid tissue reaction:** basal cell damage with massive mononuclear infiltration of the papillary dermis.
- **Epidemiology:** cutaneous LP in **up to 1% of adults;** oral lesions in **up to 4%.**
- **Pathogenesis:** an **autoimmune reaction** against basal keratinocytes expressing **altered self-antigens.**

Lesion and distribution

- **Primary lesion:** a **small, polygonal, violaceous, flat-topped papule**; lesions are separate or **coalesce into plaques that may ulcerate**, especially on **mucosa**.
- **Sites:** **flexor surfaces of the limbs**, trunk, **lumbosacral area**, **oral mucosa**, **glans penis**.
- **Male genital patterns** (eFig. 30.12):
 - Isolated or grouped papules
 - **White reticular pattern**
 - **Annular** arrangement, with or without ulceration
- **Koebner phenomenon:** **linear lesions along lines of skin trauma** (also seen in **psoriasis**).
- **Female genital pattern:** **painful erosion of erythematous plaques**; in long-standing vulvar LP, **white hyperhydrated hyperkeratosis** surrounds shallow erosions.
- **Concomitant oral LP** (buccal mucosa, tongue) is **commoner in females**.

Differential diagnosis and biopsy

- **Differential diagnosis:**
 - **Invasive and in situ SCC**
 - **Zoon balanitis**
 - **Psoriasis**
 - **Secondary syphilis**
 - **Herpes**
 - **Extramammary Paget disease (EPD)**
 - **Lupus erythematosus**
- **Biopsy** is especially needed when lesions are **small, multiple and ulcerated**.
- **Lichenoid reactions** also follow **ingested drugs and contact allergens**, so **search for an offending agent**.

Natural history and symptoms

- **Papular LP is benign:** **spontaneous resolution in up to two-thirds by 1 year**.
- **Oral and erosive forms persist longer**; **isolated SCC** has arisen in **chronic genital LP**, so perform **regular skin checks throughout the disease**.
- **Symptoms:** **pruritus is commoner in males**; **pain or burning is commoner in females**.
- **Asymptomatic genital lesions need no treatment**.
- **Vaginal erosive LP** can cause **adhesions and permanent damage**, so **manage it with care**.

TREATMENT OF LICHEN PLANUS

Line	Agents	Evidence
First-line (symptomatic genital LP)	Ultrapotent topical corticosteroid: clobetasol 0.05% or halobetasol 0.05%	Widely accepted
Adjunct	Topical calcineurin inhibitors (pimecrolimus cream, tacrolimus ointment)	Role established
Severe disease	Systemic corticosteroids	Shorten the time to clearance
Severe disease, other	Cyclosporine, tacrolimus, griseofulvin, metronidazole, acitretin	Randomised evidence generally lacking

- A **meta-analysis** found **no reliable evidence for any single treatment of erosive mucosal LP**, including ultrapotent topical steroid.

- **Set expectations: outcomes may be disappointing** for erosive LP.

Concept — LP is an **interface attack on basal keratinocytes**: the **papular skin form burns out**, but the **erosive mucosal form persists**, **scars and carries SCC risk** — which is why it is treated with the strongest topical steroid and followed long term.

Lichen Nitidus

- **Lesion**: tiny, discrete, flesh-coloured papules in large clusters.
- **Relationship to LP**: debated as a variant, but the two are **histologically distinct**; lichen nitidus (LN) shows a dense, well-circumscribed lymphohistiocytic infiltrate closely apposed to the epidermis.
- **Sites**: flexor aspects of the upper limbs, genitalia, trunk, dorsal hands; nail involvement is common.
- **Natural history**: spontaneous resolution, with the majority clearing within 1 year.
- **Management**:
 - Reassure that the lesions are **not infectious** and will resolve.
 - For pruritus: mid- to low-potency topical corticosteroids plus oral antihistamines.

Lichen Sclerosus

Epidemiology

- **Lichen sclerosus et atrophicus (LS)** is a **chronic inflammatory disease of unknown cause** with a **predilection for the external genitalia**.
- It is **6–10 times more prevalent in females**, around **menopause** or in the **prepubertal years**.
- In males it tends to affect the **older man (>60 years)**, and may cause **pain on voiding or erection**.
- There is a **strong familial predisposition**, suggesting a **genetic contribution**.
- **15–20%** of patients with genital LS have **extragenital disease**.

Clinical features

- LS is a **scarring disorder** with **tissue pallor**, **loss of architecture from fibrosis**, and **hyperkeratosis** (Fig. 30.13).
- **Purpura and fissuring** may be prominent; purpura can be severe enough to **obscure the typical white colour**.
- **Male distribution**: glans and foreskin; **perianal involvement is usually absent**, in contrast to females.
- **Male sequence**: preputial scarring → **phimosis** → **circumcision (usually curative)**; **recurrence in the circumcision scar** may occur.
- **Late stage = balanitis xerotica obliterans (BXO)**: it can involve the **penile urethra** and cause **troublesome urethral strictures**.
- **Female sequelae**: **vulvar adhesions**, **labial fusion**, **clitoral phimosis**, **vaginal obstruction**; symptoms include **itching**, **burning**, **pain** and **dyspareunia**.

Contrast with LP and LN, and malignant potential

- Despite the similar name, LS shares **little with LP and LN beyond pruritus and genital predilection**.
- **Critical distinction**: LS is associated with **SCC of the penis and vulva**, particularly **human papillomavirus (HPV)-independent variants**; LS may be premalignant.

Histology and biopsy

- **Histological features**:
 - **Basal cell vacuolation**
 - **Epidermal atrophy**

- **Marked dermal oedema**
- **Collagen homogenisation**
- **Focal perivascular papillary-dermal infiltrate**
- **Plugging of follicular and eccrine ostia**
- **Biopsy is worthwhile to confirm the diagnosis and exclude malignant change.**
- **Differential diagnosis:** erosive LP, vulvar eczema, lichen simplex chronicus, child sexual abuse, morphea, mucous membrane pemphigoid.
- **Prognostic markers:** p53, survivin, telomerase, Ki-67 and cyclin D1 may distinguish indolent LS from LS with true malignant potential, and may in future be tested routinely on biopsy specimens.

Management

- **Long-term follow-up** is required because of the SCC association; **monitor non-healing fissures, ulcers, papules and nodules** carefully.
- **Topical steroid sequence (established in females):** potent topical steroid (clobetasol propionate 0.05% or halobetasol 0.05%) for 3 months → taper → **limited twice-weekly maintenance**; this improves symptoms and may reverse the disease process.
 - **This regimen is deliberately contrary to the usual rule** of avoiding long steroid courses on genital skin.
 - **Adult males:** not definitively confirmed; **paediatric benefit** demonstrated.
- **Topical calcineurin inhibitors** help symptoms, but **comparative trials favour topical steroids**.
- **Refractory disease:** topical or systemic retinoids, photodynamic therapy.
- **Recurrence 40–50%** after apparently successful therapy; some experts therefore advise **proactive maintenance** with mid-potency steroid (mometasone furoate 0.1%) or a topical calcineurin inhibitor.
- **Vaginal lasers** have shown benefit in recalcitrant cases.

THE THREE LICHENS COMPARED

Feature	Lichen planus	Lichen nitidus	Lichen sclerosus
Nature	Idiopathic lichenoid; autoimmune against basal keratinocytes	Inflammatory; debated LP variant	Chronic inflammatory, cause unknown; scarring
Primary lesion	Polygonal violaceous flat-topped papule ; white reticular or annular	Tiny flesh-coloured papules in large clusters	White (pallor) , atrophic, hyperkeratotic plaques; purpura, fissures
Histology	Basal damage + massive papillary mononuclear infiltrate	Well-circumscribed lymphohistiocytic infiltrate apposed to epidermis	Basal vacuolation, atrophy, dermal oedema, collagen homogenisation, follicular plugging
Natural history	Papular form resolves (two-thirds by 1 year); erosive form persists	Resolves , mostly within 1 year	Chronic, relapsing; phimosis, BXO, strictures
Malignancy	Isolated SCC in chronic genital LP	None	Associated with HPV-independent penile and vulvar SCC; premalignant
First-line	Ultrapotent topical steroid	Reassurance ; mid- to low-potency steroid	Potent topical steroid for 3 months → twice-weekly maintenance

TOPICAL CORTICOSTEROID STRATEGY ACROSS GENITAL DERMATOSES

Disease	Potency	Duration
Atopic dermatitis	As needed	Short courses, rapid taper

Disease	Potency	Duration
Genital psoriasis	Low (triamcinolone for thick plaque)	≤2 weeks continuous
Reactive arthritis (genital)	Low	Short
Lichen planus	Ultrapotent	Until controlled
Lichen nitidus	Mid to low	Symptomatic
Lichen sclerosus	Potent	3 months, then twice-weekly maintenance
Seborrhoeic dermatitis	Low (hydrocortisone 2.5%)	Initial phase of severe disease only; not primary treatment

- **Pattern:** steroid strength tracks the **scarring and malignant stakes** of the disease; **LS is the single exception** to the short-course rule.

Concept — *LS is a scarring, premalignant disease, so its management is inverted — prolonged potent steroid, indefinite maintenance, biopsy and lifelong surveillance — the opposite of the minimal-steroid policy for other genital dermatoses.*

Fixed Drug Eruption

Timing and sites

- Fixed drug eruption (FDE) is **one of the commonest cutaneous drug reactions**, following **oral medications**, usually **1–2 weeks after first exposure**.
- **Sites:** lips, face, hands, feet and genitalia, **particularly the glans penis** (Fig. 30.14).
- **Genital bullous FDE is commoner in males.**
- **Re-exposure → recurrence at exactly the same site**, usually **within 24 hours** — the origin of the term "fixed".

Causative drugs

- Commonest: **sulfonamides, NSAIDs, barbiturates, tetracyclines, carbamazepine, phenolphthalein, oral contraceptives, salicylates.**
- **Urological agents reported in isolated cases: finasteride, tadalafil, fluconazole** (for vulvovaginal candidiasis).

Appearance and outcome

- On the shaft or glans: an **asymptomatic, solitary, violaceous, well-defined patch or plaque** with an **erythematous halo**; it may become **bullous, pruritic, erosive and painful**.
- **Genital differential diagnosis:** herpes simplex, insect bite.
- **Outcome:** withdrawal of the drug → **resolution**; **post-inflammatory brown pigmentation** may remain, but there is **no lasting functional defect**.
- **Histology (Fig. 30.14D):** normal stratum corneum; chronic superficial dermal change with an **eosinophilic infiltrate** — **necrotic keratinocytes, papillary dermal fibrosis, eosinophils and perivascular melanin.**

DRUG-INDUCED ERUPTIONS COMPARED: SJS VERSUS FIXED DRUG ERUPTION

Feature	SJS	Fixed drug eruption
Shared culprits	Sulfonamides, NSAIDs, tetracyclines	Sulfonamides, NSAIDs, tetracyclines
Other culprits	Penicillins, cephalosporins, anticonvulsants	Barbiturates, carbamazepine, phenolphthalein, oral contraceptives, salicylates; finasteride, tadalafil, fluconazole

Feature	SJS	Fixed drug eruption
Extent	Widespread , mucocutaneous necrosis	Solitary , same site each time
Systemic illness	Prodrome; mortality	None
Sequelae	Scarring strictures and synechiae	Pigmentation only
Treatment	Drug withdrawal + burn care	Drug withdrawal

Concept — The same drug classes can cause a trivial **fixed, solitary, recurring-in-place plaque** or a lethal **generalised necrolysis**; the **extent and systemic illness**, not the culprit drug, separate FDE from SJS.

Seborrhoeic Dermatitis

Lesion and distribution

- **Sharply demarcated pink-yellow to red-brown plaques with adherent flaky scale**; SD overlaps with eczematous dermatitis and could be grouped with it.
- **Dandruff is mild SD of the scalp.**
- SD favours **sebaceous-rich areas**, appearing in the **first few months of life** or **after puberty**, when sebaceous glands are active.
- **Sites**: scalp, eyebrows, **nasolabial folds**, ears, chest; also the **anus, glans penis and pubic area**.
- **Circumcision is somewhat protective**: penile SD risk is **higher in the uncircumcised**.

Associations

- **Adult SD** follows a **chronic relapsing course**.
- It is **particularly common in Parkinson disease** and in **acquired immunodeficiency syndrome (AIDS)** — up to **83%** of patients with AIDS.
- In the immunosuppressed it may cover a **large body surface area**.
- **Extensive or severe SD** should raise concern for underlying HIV infection or Parkinson disease.

Aetiology

- A **possible autoimmune response** to the yeast *Malassezia furfur* (*Pityrosporum ovale*) is proposed but **controversial**:
 - Organism counts in SD lesions are **only modestly higher** than in normal skin.
 - HIV patients with SD **carry no more organisms** than HIV patients without it.
- **Elevated skin-surface triglycerides and cholesterol** are another possible factor.

SEBORRHOEIC DERMATITIS VERSUS PSORIASIS

Feature	Seborrhoeic dermatitis	Psoriasis
Scale	Thinner , adherent, flaky	Thick silvery-white
Colour	Pink-yellow to red-brown	Erythematous
Nails	Rarely involved	Involved
Distribution	Sebaceous-rich: scalp, face, chest, genitals	Extensor surfaces, scalp, umbilicus, folds
Systemic clue	HIV, Parkinson disease	Family history; streptococcal trigger

Treatment

- **Mainstay on the body**: topical antifungal creams or foams (e.g. ketoconazole), with a **high response rate**.

- **Hair-bearing areas:** antidandruff shampoos containing **zinc, salicylic acid, selenium sulfide, tar, ciclopirox olamine or 1–2% ketoconazole**.
- The disease is chronic and relapsing, so treatment is **repetitive and prolonged**.
- **Low-potency steroid (hydrocortisone 2.5%)** may be used **initially in severe cases**, but **never as the primary treatment**, because of local adverse effects.

Concept — SD is a **sebaceous-territory disease tied to *Malassezia***, so **antifungal therapy, not steroid, is the backbone**; its **severity is a window onto systemic disease** — HIV and Parkinson disease.

NONINFECTIOUS ULCERS

- **Genital ulcers** have **infectious and noninfectious causes** (Box 30.4).

BOX 30.4 DIFFERENTIAL DIAGNOSIS OF ULCERS (MARGOLIS, 2002)

- **Sexually transmitted infections:** syphilis, chancroid, herpes simplex, granuloma inguinale, lymphogranuloma venereum.
- **Immune-mediated and inflammatory:** aphthous ulcer, Behçet disease, Crohn disease, pyoderma gangrenosum.
- **Vasculitic:** granulomatosis with polyangiitis (Wegener granulomatosis), leucocytoclastic vasculitis.
- **Traumatic and self-inflicted:** genital bite wound, factitial dermatitis.
- **What the box is for:** a genital ulcer is **infectious until proven otherwise**; every noninfectious entity on the list is a **diagnosis reached only after sexually transmitted infection has been excluded**, and the traumatic causes are found **only by history**.

Aphthous Ulcers and Behçet Disease

Aphthous ulcers

- **Aphthous ulcers** are **small, painful erosions**, commonly of the **oral cavity (canker sores)** and **occasionally the genitalia**.
- **Coexisting oral and genital aphthous ulcers** should prompt serious consideration of **Behçet disease (BD)**.

Nature and pathogenesis

- **BD** is a **generalised relapsing and remitting ulcerative mucocutaneous disease** with a **genetic predisposition** and an **autoimmune mode of pathogenesis**.
- Many loci are implicated; the **strongest genetic association is HLA-B51**.
- **Oxidative stress** is implicated: **neutrophils overproduce superoxide radicals**.
- Other proposed mechanisms include **interleukin-10 (IL-10) gene mutations**.
- **Treatment response varies widely**, which implies that the **inflammatory pathways are not uniform** across patients.

EPIDEMIOLOGY OF BEHÇET DISEASE

Region	Prevalence per 100,000
Turkey	80 (highest)
Israel	15
Japan	10–12
United States	0.12–5.0 (rare)

- **Sex:** frequency is roughly **equal**, but **males run a more severe course**.

Clinical features

- **Systemic associations:**
 - Epididymitis
 - Thrombophlebitis
 - Aneurysm, particularly of the pulmonary artery
 - Gastrointestinal, neurological and arthritic disease
- **Classic triad: oral ulcers, genital ulcers (Fig. 30.19) and ocular involvement (uveitis).**
- **Genital versus oral ulcers: genital lesions are larger and more painful than oral lesions.**
- **Ocular involvement is common and may cause blindness.**

INTERNATIONAL CRITERIA FOR BEHÇET'S DISEASE (2014) — DIAGNOSIS REQUIRES A SCORE OF AT LEAST 4 POINTS

Feature	Points
Ocular lesions	2
Oral aphthosis	2
Genital aphthosis	2
Skin lesions	1
Central nervous system involvement	1
Vascular manifestations	1
Positive pathergy test (optional)	1

- **Pattern:** the three mucosal and ocular features score double — **any two of them alone reach the threshold.**
- **Sensitivity versus specificity:** oral ulceration is the most sensitive lesion; genital ulceration is the most specific, and therefore the **most clinically useful lesion** for diagnosis.
- **Exclude first:** simple or complex aphthosis, **primary syphilis, herpes simplex, chancroid.**
- There is **no confirmatory laboratory, radiological, genetic or histological test**; the diagnosis rests **exclusively on the aggregate clinical findings.**

Treatment

- The course is **protean**, and **randomised controlled trials are limited.**
- **Agents used, with variable success:**
 - Corticosteroids
 - **Dapsone**
 - **Colchicine**
 - Immunosuppressants
 - **5-aminosalicylic acid (5-ASA) derivatives**
 - **Cyclosporine A**
 - **TNF- α inhibitors, especially infliximab and adalimumab**
- **Earlier and more aggressive immunosuppressive and biologic treatment** of significant organ involvement **has improved overall outcome.**
- **Request rheumatology consultation** when BD is suspected.

Concept — BD is a **clinical diagnosis by aggregation**. No test confirms it, so it rests on **genital ulceration (the most specific sign)** combined with oral and ocular disease, and only after the infective ulcers have been excluded.

Pyoderma Gangrenosum

Definition and associations

- **Pyoderma gangrenosum (PG)** is a rare ulcerative neutrophilic dermatosis associated with systemic illness:
 - Inflammatory bowel disease
 - Arthritis
 - Collagen vascular disease
 - Chronic active hepatitis
 - HIV infection
 - Myeloproliferative disorders
- It most commonly affects females in the second to fifth decades; the pathogenesis is likely autoimmune, given its association with other autoimmune diseases.

Epidemiology

- Genital PG is rare; 20–50% of cases are idiopathic; annual incidence in the United States is about 1 per 100,000.

Morphology

- Painful cutaneous and mucosal ulceration with extensive tissue loss and a purulent base (Fig. 30.20).
- Classic appearance: a rapidly developing lower-leg ulcer with an overhanging, undermined violaceous border, arising at a site of trauma.
- Unusual sites: penis, scrotum, vulva, peristomal skin.

Diagnosis

- As in BD, there is no pathognomonic test or histological feature; a history of underlying systemic disease raises suspicion.
- Biopsy 5–10 mm from the ulcer edge, where a dense neutrophilic infiltrate is found.
- Differential diagnosis of penile PG, beyond the ulcerative sexually transmitted infections:
 - Calciphylaxis
 - BD
 - Necrotising fasciitis and Fournier gangrene (FG)
 - Cutaneous metastatic Crohn disease
 - Deep fungal infection
 - Pemphigus vegetans
 - Neoplasia
 - Erosive LP
 - Trauma, factitious damage

TREATMENT OF PYODERMA GANGRENOSUM

Line	Agents
Core	Local plus systemic corticosteroids
Adjunctive immunosuppression	Cyclosporine, TNF- α inhibitors, IVIG
Steroid combinations (small case numbers)	Minocycline, sulfasalazine, thalidomide
Genital PG, topical	Topical calcineurin inhibitors

BEHÇET DISEASE VERSUS PYODERMA GANGRENOSUM

Feature	Behçet disease	Pyoderma gangrenosum
Pathology	Autoimmune; neutrophil oxidative stress; HLA-B51	Neutrophilic dermatosis ; autoimmune
Demography	Turkey, Middle East, Japan ; equal sex ratio, males more severe	Females, second to fifth decades
Ulcer	Aphthous , oral and genital; genital larger and more painful	Large, purulent, undermined violaceous border ; arises at trauma
Systemic links	Eye (uveitis) , vascular (pulmonary aneurysm, thrombophlebitis), epididymitis, CNS	IBD, arthritis, myeloproliferative disease, hepatitis, HIV
Confirmation	Points-based clinical criteria ; no test	Edge biopsy (neutrophilic infiltrate); no pathognomonic feature
Treatment	Colchicine, dapsone, steroids, anti-TNF ; rheumatology	Systemic steroid ± cyclosporine, anti-TNF, IVIG

Figure notes

- **Fig. 30.20** PG of the genitals in a female patient with **rheumatoid arthritis** (A), and of the penis and scrotum (B) (Pinard et al., 2018; Callen et al., 1993).

Traumatic Causes

- **Local trauma belongs in every differential** of genital lesions, including ulcers.

ACCIDENTAL VERSUS SELF-INFLICTED GENITAL TRAUMA

Category	Forms	Features / associations
Accidental ("innocent trauma")	Sexual practices (including genital bite wounds); ornamentation (piercing); unusual hygiene practices (cleaning)	Needs a thorough sexual history and a sexual-abuse screen
Factitial dermatitis (dermatitis artefacta)	Self-inflicted psychocutaneous disorder	Usually an unconscious motive or mental illness ; occasionally deliberate for secondary gain (e.g. product-liability litigation); associated with borderline personality disorder
Psychogenic (neurotic) excoriation	Self-inflicted picking (Fig. 30.21)	Underlying depression, bipolar disorder, delusional disorder, obsessive-compulsive disorder

- **Other diagnoses to consider:** **Munchausen syndrome** (and **Munchausen syndrome by proxy** where a dependant is the victim), **body dysmorphic disorder**, and **malinger** if secondary gain exists.
- **Recognition clue:** **oddly configured, angulated erosions and ulcers** should always prompt consideration of factitial disease or neurotic excoriation.

Concept — *Noninfectious genital ulcers are **diagnoses of exclusion and of context**: **BD and PG have no confirmatory test**, and trauma is revealed only by history, so **systemic associations and ulcer shape** carry the diagnosis once infection has been excluded.*

INFECTIONS AND INFESTATIONS**Sexually Transmitted Diseases**

- Many sexually transmitted diseases have cutaneous manifestations (Fig. 30.22): **herpes simplex**, **molluscum contagiosum**, **syphilis**, **lymphogranuloma venereum**, **granuloma inguinale**, **chancroid** and **HPV** (condyloma acuminata).
- These are covered in **Chapter 29**.

Balanitis and Balanoposthitis

Definitions and causes

- **Balanitis** is inflammation of the **glans**; **balanoposthitis** is inflammation of the **glans and prepuce**, so it occurs in the **uncircumcised**.
- **Causes by age:**
 - **Children:** bacterial infection predominates.
 - **Adults:** poor hygiene, trauma, irritant contact dermatitis, and bacterial, fungal (candidal, Fig. 30.23) or viral infection.
- **Presentation:** usually penile pain and erythema.

Treatment

- **Sequence:** remove irritants → improve hygiene → topical antibiotics and antifungals → occasionally a short course of low-potency topical corticosteroid.
- **If initial treatment fails:**
 - Widen the differential to **neoplasia**, **Zoon balanitis**, **psoriasis**, and **alternative infections** (HPV, HSV).
 - **Consider biopsy.**
- Balanoposthitis tends to occur with **phimosis**; **circumcision** may be curative in selected recurrent cases.
- **Bacterial superinfection** also occurs with **poor hygiene** and **neutropenia**.

Cellulitis and Erysipelas

CELLULITIS VERSUS ERYSIPELAS

Feature	Cellulitis	Erysipelas
Depth	Deep dermis and subcutaneous tissue	Upper dermis and superficial lymphatics
Organism	Gram-positive: β -haemolytic streptococci and <i>S. aureus</i>	Usually <i>Streptococcus pyogenes</i>
Border	Indistinct	Distinct, sharply demarcated
Typical patient / site	Genital skin (Fig. 30.24); mixed flora in diabetes	Extremes of age; often the face

Cellulitis in detail

- **Entry:** usually through a **break in the skin barrier**; in the immunocompromised, haematogenous spread is more common.
- **Systemic signs:** fever, chills, malaise.
- **Local signs:** **rubor, calor, dolor, tumor** (redness, warmth, pain, swelling); **petechiae, haemorrhage or superficial bullae** may occur.
- **Treatment:**
 - **Systemic antibiotics** active against β -haemolytic streptococci and *S. aureus*.
 - **Methicillin-resistant *S. aureus* (MRSA)** cover if risk factors are present.
 - **Cultures are difficult** to obtain, so rely on **local antimicrobial sensitivity patterns**.
 - In **diabetes**, flora may be mixed, so **broaden cover**.
 - **Mark the border at the start of therapy** to judge progression or resolution.

Fournier Gangrene (Necrotising Fasciitis of the Perineum)

Microbiology and triggers

- **Fournier gangrene (FG)** is a **potentially life-threatening progressive infection** of the perineum and genitalia.
- **Microbiology: polymicrobial** (gram-positive, gram-negative and anaerobic); common pathogens are *Escherichia coli*, *Bacteroides* spp., *S. pyogenes* and **S. aureus*; *fungus FG** has been reported.
- **Inciting events:**
 - Skin or mucosal tear
 - **Penetrating trauma**
 - **Recent urogenital or colorectal instrumentation**
- **Risk factors:**
 - **Alcoholism**
 - **Diabetes**
 - Cancer, malnutrition
 - **Chronic kidney disease**
 - **Urethral stricture disease**
 - Advanced age
 - Genitourinary infection
 - **Peripheral vascular disease**

Progression and spread

- **Hallmark sequence:** cellulitis (erythema, swelling, pain) → blisters → visible ischaemia → foul-smelling necrosis (Fig. 30.25).
- **Spread along fascial planes** from the genitalia and perineum to the **abdominal wall fascia (Scarpa's)**; the skin shows **only a small fraction** of the infected and necrotic tissue.
- **FG is a surgical emergency:** progression takes **hours**, and **septicaemia is the usual cause of death**.
- **Consider FG in every genital soft-tissue infection consultation.**

Clinical clues to FG

- **Pain out of proportion** to the visible infection.
- A **greyish cast** to the skin.
- A **rank odour**, uncharacteristic of uncomplicated cellulitis.

Imaging

- Plain radiographs, **computed tomography (CT)** and bedside ultrasonography may show **soft-tissue gas**; **CT has the highest specificity**.
- **Imaging must not delay surgery in obvious cases.**

Management sequence

- **Broad-spectrum antibiotics + extensive debridement to healthy bleeding tissue → second-look operation at 24–48 hours → repeated debridements** as needed → **multidisciplinary care** (urology, general surgery, infectious diseases, critical care).
- The **testes and other contents of the tunica vaginalis are usually spared** (Fig. 30.25B); **abdominal wall loss may be extensive** because of fascial spread.
- **Adjuncts:** **hyperbaric oxygen** is **controversial**, though some report favourable results; **vacuum-assisted closure** may be beneficial.

Outcome and prognostic scores

- **Mortality 7.5–40%** despite aggressive modern management.

PROGNOSTIC SCORES IN FOURNIER GANGRENE

Score	Scope
Fournier's Gangrene Severity Index (FGSI)	FG-specific
Uludağ FGSI	FG-specific
Age-Adjusted Charlson Comorbidity Index (ACCI)	General comorbidity
Surgical Apgar Score (SAS)	General operative

- **All four are validated** for FG; adopting one assists therapeutic decisions.
- **Survivors:**
 - **Disability and reduced function for months to years**
 - **Sexual dysfunction is common (~65%)**
 - **Long-term multispecialty care** is needed

Concept — FG is a *fascial, not a skin, disease*: the skin shows only a fraction of the necrosis, so **suspicion (pain out of proportion)**, **not imaging**, triggers **immediate radical debridement**, and the testes survive within their own tunica compartment.

Folliculitis*Features*

- **Inflammation of the hair follicle**, producing **perifollicular pustules or papules on an erythematous base**.
- **Sites**: heavily hair-bearing areas (scalp, beard, axilla, **groin**, buttocks).
- **Aggravated by**: occlusion (e.g. truck drivers), shaving, rubbing, clothing irritation.
- **Symptoms**: pruritus or pain, or none.
- **Cause**: usually **bacterial infection** of the superficial or deep follicle; also fungal, viral or noninfectious.
- **Diagnosis**: **cultures are generally negative**, so the diagnosis is **clinical**.

Hot-tub folliculitis

- Follows **contaminated hot tubs and swimming pools**; the organism is usually *Pseudomonas aeruginosa* (Fig. 30.26).
- **Onset within 48 hours** of exposure.
- **Pool and hot-tub *P. aeruginosa* isolates are overwhelmingly multidrug-resistant**, which matters most for the immunosuppressed.

Treatment

- **Hygiene, removal of irritants, and topical or systemic antiviral, antibiotic or antifungal agents.**
- **Recurrence is common.**
- **Failure of conservative measures → lesion culture with sensitivity testing.**

Furunculosis*Features*

- A deeper follicular infection forming a **red, tender boil or abscess**.
- **By definition a furuncle arises from a hair follicle**, whereas abscesses can occur anywhere.
- **Organism**: *S. aureus*.
- **Sites**: areas of minor trauma — **groin, axilla, buttocks** (Fig. 30.27).

- **Risk factors:**
 - Impaired immunity
 - Skin trauma
 - **Obesity**
 - Poor hygiene
 - **Diabetes**
- **Differential diagnosis:** epidermal cyst, Bartholin gland duct cyst.

Treatment

- **Warm compresses; incision and drainage** for larger lesions.
- **Associated cellulitis** → systemic antistaphylococcal antibiotic, with MRSA cover guided by community prevalence.
- **Recurrence prevention:** nasal mupirocin to eliminate carriage.

SOFT-TISSUE INFECTIONS BY DEPTH: FOLLICULITIS TO FOURNIER GANGRENE

Entity	Level	Organism	Key sign	Management
Folliculitis	Superficial or deep follicle	Bacterial (<i>S. aureus</i>); hot tub → <i>P. aeruginosa</i>	Perifollicular pustules	Hygiene; topical or systemic agent
Furunculosis	Deep follicle → abscess	<i>S. aureus</i>	Red, tender boil	Warm compress, incision and drainage, anti-staphylococcal ± MRSA, nasal mupirocin
Erysipelas	Upper dermis + lymphatics	<i>S. pyogenes</i>	Sharp border	—
Cellulitis	Deep dermis + subcutis	β-haemolytic streptococci, <i>S. aureus</i> ; mixed in diabetes	Indistinct border	Systemic antibiotics; mark the border
Fournier gangrene	Fascia	Polymicrobial	Pain out of proportion, soft-tissue gas, grey skin, odour	Emergency debridement + broad-spectrum antibiotics; second look at 24–48 hours

Concept — Depth decides severity: moving down from follicle to fascia, the organism set broadens, the border blurs, and management escalates from hygiene to emergency surgery.

Hidradenitis Suppurativa (Acne Inversa)

Definition and epidemiology

- **Hidradenitis suppurativa (HS)** is a chronic inflammatory disease of apocrine gland-bearing skin; the primary sites are the intertriginous axillae and anogenital regions.
- **Onset** is after puberty, usually in the second or third decade.
- It is **commoner in females** and in patients of Black ancestry.
- **Genetic predisposition** is likely: up to 41% have a family history.

Pathogenesis

- Originally considered an apocrine disease, HS is **now considered follicle-centred**.
- **Sequence:** follicular plugging and swelling → rupture of follicular contents (bacteria, keratin) into the dermis → marked inflammation → abscesses and sinus tracts.

Clinical features and complications

- **Features:** painful inflammatory nodules, skin tunnels and sterile abscesses in the axillae, groin, perianal and inframammary areas (Fig. 30.28).
- Over time: draining sinus tracts and hypertrophic scars.
- **Serious complications:**
 - Hypoproteinaemia
 - Secondary amyloidosis
 - Fistulae to the urethra, bladder, peritoneum and rectum
 - Lymphatic obstruction
 - Strictures and contractures
 - SCC in heavily scarred areas

General measures

- Patient education, smoking cessation, hygiene, weight reduction.
- Management of metabolic syndrome and hormone-related disorders.
- Reduce friction and moisture (loose undergarments, absorbent powder).
- No single intervention is universally effective.

MEDICAL AND SURGICAL TREATMENT OF HIDRADENITIS SUPPURATIVA

Category	Agent	Dose	Note
Topical antibiotic	Clindamycin	—	Oral tetracycline was no better than topical clindamycin in a double-blind RCT
Oral antibiotic combinations	Clindamycin + rifampicin; minocycline + rifampicin	—	Beneficial for some
Anti-inflammatory	Dapsone	50–200 mg/day	—
Mineral	Zinc	40–80 mg/day elemental	—
Retinoids	Acitretin; isotretinoin	25–50 mg/day; 1 mg/kg/day	—
Immunosuppressant	Cyclosporine	4 mg/kg/day	—
Metabolic	Metformin	500 mg/day	—
Hormonal (females)	Spirolactone, oral contraceptives	—	—
Hormonal (males)	Finasteride, dutasteride	—	—
Systemic steroid	Corticosteroids	—	Improve HS, but relapse is the rule on cessation
Biologic	Adalimumab	40 mg subcutaneously weekly	Effective
Avoid	Lithium	—	Exacerbates HS or blunts treatment response

Category	Agent	Dose	Note
Surgery	Unroofing of inflamed nodules and abscesses; wide excision with skin grafting for chronic or extensive disease; CO₂ laser excision	—	Recurrent incision and drainage is discouraged

- **Pattern:** medical therapy targets **infection, inflammation and hormones**; **only definitive excision removes the diseased follicular unit**, whereas incision and drainage merely recurs.

Figure notes

- **Fig. 30.28 HS:** (A) **painful papules and draining sinus tracts**; (B) histology — **follicular plugging connected to a dilated apocrine duct**; (C, D) **severe genital involvement**, which makes surgery difficult (Danby and Margesson, 2010; Bologna et al., 2003).

Concept — HS is a **follicular occlusion disease**: plugged follicles rupture and tunnel, so medical therapy only damps the inflammation, while **unroofing or wide excision removes the tract** that repeated drainage leaves behind.

Corynebacterial Infection (Trichomycosis Axillaris and Erythrasma)

TRICHOMYCOSIS VERSUS ERYTHRASMA

Feature	Trichomycosis	Erythrasma
Organism	<i>Corynebacterium</i> spp.	<i>Corynebacterium minutissimum</i>
Target	Hair shaft: axillary (trichomycosis axillaris) or pubic (trichomycosis pubis)	Skin of moist areas: groin, axilla; macerated plaque between the toes
Lesion	Yellow, red or black nodules on hair shafts (Fig. 30.29A) + foul odour	Well-demarcated, light red to dark brown scaling patch
Symptoms	—	Pruritic or asymptomatic
Risk factors	Poor hygiene, hyperhidrosis, obesity, high humidity	—
Mimics	Pediculosis pubis, piedra (fungal) — distinguished by magnification	Tinea cruris
Key test	Magnification	Wood's lamp: bright coral-red fluorescence (Fig. 30.29C)
Treatment	Shaving, trimming, hygiene (immediate improvement); antibacterial soaps prevent recurrence; pubic: clindamycin gel, bacitracin, oral erythromycin	Antibacterial soaps; aluminium chloride 10–20%; clindamycin 1% solution or gel; miconazole 1% cream; oral erythromycin 500–1000 mg/day for severe cases

Ecthyma Gangrenosum

Definition and lesion

- The **cutaneous manifestation of gram-negative bacteraemia**, most commonly *P. aeruginosa*, in **debilitated, immunocompromised patients**.
- **Sites:** perineum, buttocks, axillae, lower limbs.
- **Lesion sequence:** tender grouped erythematous or purpuric macules → **bullae** or a **haemorrhagic pustule** → **rupture** → **gangrenous ulcer with a thick black eschar** (Fig. 30.30).

Diagnosis

- **Biopsy is diagnostic:** **necrotising vasculitis with gram-negative organisms** — **necrotic vessels surrounded by a "blue haze" of organisms**, with little inflammation.

- **Differential diagnosis:**

- PG
- **Necrotising vasculitis**
- **Cryoglobulinaemia**
- **Calciophylaxis**
- **Septic emboli** from other organisms: *Candida*, *Aspergillus*, *Citrobacter*, *E. coli*, *Aeromonas hydrophila*, *Fusarium*

Management and prognosis

- The prognosis is **poor**, reflecting the underlying sepsis.
- **Immediate intravenous antipseudomonal antibiotics, with debridement if needed.**

Genital Bite Wounds

- **Organism:** after deliberate or accidental genital bites, *Eikenella corrodens* (normal human oral flora) is implanted in the skin.
- **Result:** **rapid, extremely painful necrotic ulceration** at the bite site (Fig. 30.31); **progression to necrotising fasciitis** has been described.
- **Distinguishing triad** from sexually transmitted and other ulcers:
 - **Rapidity of ulceration**
 - **Extraordinary pain**
 - **History of traumatic orogenital contact**
- **Treatment:** high-dose oral amoxicillin-clavulanate 875/125 mg twice daily until healed, plus aggressive local wound care.
- **Surgery is often needed:** irrigation → debridement → primary closure → occasionally reconstruction.

Candidal Intertrigo

Features

- **Candidal infection of macerated skin folds** in warm, moist areas: groin, axillae, **beneath the breasts**, intertriginous zones.
- **Pruritic, red skin** with characteristic **satellite lesions** (Fig. 30.32).
- **Differential diagnosis:** tinea cruris, pemphigoid, psoriasis, SD, contact dermatitis.
- **Risk factors:** obesity, diabetes, immunocompromise, prolonged heat and moisture.

Diagnosis

- **KOH preparation** shows round yeast cells and elongated pseudohyphae; culture is usually unnecessary.
- Candidal intertrigo may be the presenting sign of diabetes, so test for diabetes.

Treatment

- **Daily topical imidazole** for at least 2 weeks.
- **Oral fluconazole 150 mg** is occasionally required.
- **Relapse prevention:** drying powders, loose clothing.

Dermatophyte Infection

Organisms and pathogenesis

- Dermatophytes are **filamentous fungi of three genera** — *Trichophyton*, *Microsporum*, *Epidermophyton* — that invade **keratinised tissue** (skin, hair, nails).

- **Mechanisms:**

- **Keratinases** digest keratin, and the breakdown products aid invasion.
- **Cell-wall mannans** are immunoinhibitory.

Tinea cruris ("jock itch")

- **Commonest in postpubertal males.**
- **Favoured by:** heat and humidity, concomitant *tinea pedis*, diabetes, obesity.
- **Distribution:** an erythematous rash of the inner thighs and inguinal region; the scrotum and penis are usually spared, although isolated penile dermatophytosis occurs.
- **Significant scrotal involvement → suspect cutaneous candidiasis.**
- **Lesion:** sharply demarcated erythematous plaques with a raised scaly border (Fig. 30.33); intensely pruritic, but pruritus may be absent in the immunocompromised.
- **Mimics:** SD, psoriasis, contact dermatitis, cellulitis, intertrigo, erythrasma.

Diagnosis

- **Skin scraping + KOH preparation** (branching hyphae); culture is rarely needed.
- **A negative scraping from an untreated lesion → consider an alternative diagnosis.**

Treatment

- **Prevention:** loose clothing, laundering contaminated garments, weight reduction, glycaemic control, drying powders.
- **Topical antifungals are the mainstay;** powder forms also dry the area.
- **Treat active disease only,** not the post-inflammatory hyperpigmentation of chronic recurrent infection.
- **Systemic antifungals are rarely needed;** for follicular involvement too deep for topicals, the drug of choice is **terbinafine 250 mg/day for 1 week.**
- **Topical steroids are generally not indicated.**

ANTIFUNGAL SPECTRUM

Class	Agents	Yeast (<i>Candida</i>)	Dermatophytes
Azoles (imidazoles)	e.g. miconazole, ketoconazole	Yes	Yes
Allylamines	Terbinafine, naftifine	Less effective	Yes
Polyene	Nystatin	Yes	No

- **Trap:** use an azole when yeast and dermatophyte cannot be separated — nystatin fails *tinea*, and allylamines are weaker against *Candida*.

RED GROIN RASH: DIFFERENTIAL AT A GLANCE

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Organism	<i>Candida</i>	<i>Trichophyton</i> , <i>Microsporum</i> , <i>Epidermophyton</i>	<i>C. minutissimum</i>	None
Scrotum	Involved	Usually spared	—	May be involved
Morphology	Red, macerated; satellite lesions	Raised scaly advancing border; central clearing with hyperpigmentation	Light red to brown scaling patch	Non-scaling erythematous plaque
Key test	KOH: yeast + pseudohyphae	KOH: branching hyphae	Wood's lamp: coral-red	KOH negative

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
First-line	Topical imidazole ≥ 2 weeks	Topical azole or allylamine	Antibacterial or erythromycin-based	Low-potency steroid

Concept — Intertriginous rashes share warmth, moisture and redness, so **one bedside test (KOH preparation or Wood's lamp) sorts them**; **scrotal involvement and satellite lesions point to Candida**, while **scrotal sparing with an advancing scaly border points to tinea**.

Infestation

- **Pediculosis pubis and scabies (*Sarcoptes scabiei*) are the commonest genital infestations.**

Pediculosis pubis

- Caused by the **crab louse (*Phthirus pubis*)**; it is **pruritic** and **sexually transmitted**.
- It **coexists with other sexually transmitted diseases** — affected adolescents carry a **markedly higher risk of concomitant gonorrhoea or chlamydia**, so **screen for them**.
- It also involves **other hair-bearing sites: eyelashes, beard, axillae**.
- **Diagnosis: crab lice attached to hairs, often with perifollicular erythema** (Fig. 30.34).
- **Transmission: usually sexual contact**; contaminated **clothing, bedding and towels** are occasionally implicated.
- **Standard treatment sequence: 5% permethrin cream overnight to all affected hair-bearing areas → remove all visible nits → clean underwear → repeat application at 1 week.**
 - The **second application is essential**, because a **single application frequently fails**.
- **Refractory disease, or eyelash involvement (phthiriasis palpebrarum): add oral ivermectin.**
- **Alternative: malathion 0.5% lotion.**
- **Epidemiology: incidence in industrialised countries has fallen sharply** with widespread **pubic hair removal ("Brazilian waxing")**.

Scabies — epidemiology and transmission

- Caused by the **mite *Sarcoptes scabiei***; a **worldwide problem** spread by **overcrowding, delayed treatment of index cases and poor public awareness**.
- **Highest prevalence in the Pacific and Latin American regions.**
- **Transmission: direct, prolonged skin-to-skin contact**; common among **close contacts and family members**.
- **Mite load:**
 - **Immunocompetent host: usually fewer than 100 mites.**
 - **Immunosuppressed host: far greater numbers — crusted (Norwegian) scabies.**
- **Incubation: days to months, typically about 6 weeks.**

Scabies — clinical features and diagnosis

- **Severe pruritus is the hallmark, worst at night or after bathing.**
- **Distribution: genital areas in both sexes; sides and webs of fingers; axillary folds, wrists, elbows.**
- **Lesions: small erythematous pruritic papules, and excoriations with secondary bacterial infection** (Fig. 30.35).
- **Thin grey or white serpiginous burrows are pathognomonic.**
- **Crusted scabies** on genital skin presents as **thickly crusted plaques**, as elsewhere.
- **Secondary staphylococcal or streptococcal infection** occurs.

- **Without visible burrows**, consider a broad differential: **AD, pyoderma, psoriasis, other insect bites**.
- **Diagnosis**: usually **clinical**; **skin scraping microscopy or dermoscopy** identifies **burrows, mites and eggs**.

Scabies — treatment

- **Treatment sequence**: **5% permethrin cream to the entire body overnight** → **repeat at 1 week** → **treat household members and close contacts**.

SCABICIDES

Agent	Regimen	Status
Permethrin 5% cream	Whole body overnight; repeat at 1 week	Treatment of choice
Oral ivermectin	200 µg/kg per dose, 2 doses 1–2 weeks apart	Alternative ; slightly less effective than permethrin when given as a single dose (randomised comparative trial)
Lindane	Topical	Not favoured : CNS toxicity in children and rising mite resistance

- **Pruritus** may persist for **several weeks** despite successful treatment; **treat all intimate contacts** to prevent reinfestation.
- **Postscabies nodules**: **itchy nodules** may persist on the **glans penis** after effective treatment; treat with **intralesional minute amounts of dilute triamcinolone acetonide**.

PEDICULOSIS PUBIS VERSUS SCABIES

Feature	Pediculosis pubis	Scabies
Organism	<i>Phthirus pubis</i> (crab louse)	<i>Sarcoptes scabiei</i> (mite)
Transmission	Sexual ; fomites occasionally	Prolonged skin-to-skin contact ; household
Sites	Pubic hair; eyelashes, beard, axillae	Genitals, finger webs, wrists, elbows, axillary folds
Pathognomonic sign	Lice attached to hairs , nits	Serpiginous burrows
Immunosuppressed form	—	Crusted (Norwegian) scabies
First-line	Permethrin 5% to hair-bearing areas, repeat at 1 week , nit removal	Permethrin 5% to the whole body, repeat at 1 week
Alternatives	Oral ivermectin (refractory or eyelash); malathion 0.5%	Oral ivermectin ; lindane disfavoured
Contacts	Screen for other STDs	Treat household and intimate contacts

Concept — Both infestations are cured by the **same logic**: a **second permethrin application at 1 week** kills the newly hatched generation, and **treating contacts (and screening them for STDs)** prevents reinfestation; residual itch does **not** mean treatment failure.

NEOPLASTIC CONDITIONS

Squamous Cell Carcinoma in Situ

Nomenclature and epidemiology

- **Squamous cell carcinoma in situ (SCCis)** is a **full-thickness intraepidermal carcinoma**, also called **intraepithelial neoplasia**, and named by organ: **penile intraepithelial neoplasia (PeIN)** and **vulvar intraepithelial neoplasia (VIN)**.
- **Bowen disease**: described by **Bowen in 1912**.

- On **extragenital sites** it is strongly associated with **ultraviolet exposure**.
- **Demography**: usually presents in the **seventh decade**, with a **slight female predominance**.
- **Course**: **indolent**; rarely progresses to invasion.
- **Erythroplasia of Queyrat**: SCCis on **mucosal surfaces of the male genitalia**, notably the **glans of the uncircumcised** (Fig. 30.36).

Risk factors and classification

- **HPV coinfection** in genital sites: **types 8 and 16 predominate**, with other types in the remainder.
- **Other risk factors**:
 - **Ionising radiation**
 - **Immunosuppression**
 - **Thermal injury**
 - **Arsenic**
 - **Chronic dermatoses (long-standing LP)**
 - **LS of the glans**
- **Classification**: like invasive carcinoma, **PeIN is classified as HPV-related or non-HPV-related**.

Clinical features and diagnosis

- **Lesion**: **sharply demarcated, solitary, pink-to-red scaly plaque**.
- **Mimics**: **basal cell carcinoma (BCC)**, eczema, seborrhoea, psoriasis.
- **Site variants**:
 - **Vulvar SCCis**: may be **heavily pigmented**, mimicking **melanoma** and genital warts.
 - **Penile-shaft SCCis**: **thicker, verrucoid**.
- **Symptoms**: usually **asymptomatic**; sometimes pruritic or painful.
- **Diagnosis**: **histological** — **sample several areas to exclude dermal invasion**.

TREATMENT OF SQUAMOUS CELL CARCINOMA IN SITU

Setting	Modality
Accessible site (e.g. scrotum)	Simple excision with a 5-mm margin (favoured)
Tissue preservation critical	Mohs micrographic surgery, laser, cryoablation
Selected cases	Topical 5-fluorouracil or imiquimod 5% — variable response, higher recurrence than surgery

Bowenoid Papulosis

Features and diagnosis

- **Demography**: uncommon; affects the **penis and vulva of sexually active adults**, peaking in the **third decade**; it is **sexually transmitted**.
- **Histology**: resembles Bowen disease, **but atypical keratinocytes are scattered discontinuously** through the epidermis; the **full-thickness atypia** may be mistaken for Bowen disease (Fig. 30.37B).
- **Lesion**: **small, well-demarcated erythematous (often brown) papules**; may coalesce into **verrucous plaques** resembling warts (Fig. 30.37A).
- **HPV**: associated with **HPV type 16**, and occasionally others.
- **Differential diagnosis**: psoriasis, LP, **condyloma**, **pigmented Bowen disease**.
- **Female partners** have an **increased risk of cervical neoplasia** and need **close cervical follow-up**.

Course and treatment

- In males the course is generally **benign**, and **spontaneous regression occurs**; in a **young, reliable patient**, **conservative management** is acceptable.
- If treatment is desired:
 - **Topical**: 5-fluorouracil 0.5%, tazarotene 0.5% cream, imiquimod 5%
 - **Ablative**: electrocoagulation, liquid nitrogen cryotherapy, CO₂ laser
- Relapse occurs with every modality.

Squamous Cell Carcinoma

- Invasive genital SCC (Fig. 30.38) is covered in **Chapter 82**.

Figure notes

- **Fig. 30.38 SCC**: (A) **exophytic erosive keratinising glans lesion**; (B) **atypical keratinocytes invading the dermis** (Callen et al., 1993; Elston and Ferringer, 2009).

Verrucous Carcinoma (Buschke–Lowenstein Tumour)*Features*

- A rare, low-grade variant of SCC: **slow-growing, locally aggressive, exophytic**, with **low metastatic potential**.
- The **Buschke–Lowenstein tumour** is **verrucous carcinoma of the anogenital mucosa**; it may represent **up to 24% of penile tumours**.
- **Sites**: **uncircumcised males, glans or prepuce**; also the vulva, vagina, cervix and anus.
- **HPV**: associated with **types 6 and 11** — **not the oncogenic types 16 and 18**.
- **Cofactors**: **chronic inflammation, tobacco**, other carcinogens.
- **Lesion**: **warty, "cauliflower-like"**, often **large and fungating** (Fig. 30.39); **deeply destructive** locally.
- **Extragenital sites**: **oral and nasal cavities, plantar feet**.

Treatment

- **Local excision** is preferred, but **recurrence is high**.
- **Mohs surgery** traces the tumour and minimises tissue loss.
- **Primary radiotherapy** is **relatively contraindicated**: risk of **anaplastic transformation** and **increased metastatic potential**.

SQUAMOUS INTRAEPITHELIAL AND VERRUCOUS NEOPLASIA COMPARED

Feature	SCCis / erythroplasia of Queyrat	Bowenoid papulosis	Verrucous carcinoma (Buschke–Lowenstein)
Age	Seventh decade	Third decade , sexually active	Uncircumcised males
HPV	8, 16 ; also non-HPV (LS, LP)	16	6 and 11 (not 16/18)
Lesion	Solitary red scaly plaque; glans velvety red (Queyrat)	Multiple small brown–red papules	Large cauliflower-like, fungating mass
Histology	Full-thickness continuous atypia	Discontinuous atypia (mimics Bowen)	Well-differentiated, bulbous pushing border
Behaviour	Indolent; rarely invades	Regresses spontaneously in males	Locally destructive, rarely metastasises
Partner risk	—	Cervical neoplasia in female partners	—

Feature	SCCis / erythroplasia of Queyrat	Bowenoid papulosis	Verrucous carcinoma (Buschke–Lowenstein)
Treatment	Excision (5-mm margin) , Mohs, laser, cryo; topical 5-FU or imiquimod	Conservative , or topical or ablative	Excision / Mohs; no primary radiotherapy

- **Genital HPV types map to lesions: 6/11 → condyloma and verrucous carcinoma; 16 → bowenoid papulosis and SCCis.**

Basal Cell Carcinoma

Epidemiology and subtypes

- **BCC is the commonest cutaneous neoplasm overall**, arising on **chronically sun-exposed skin** (head and neck).
- **Genital BCC is very rare**, most often on the **scrotum in males** and the **vulva in females**.
- **Subtypes: nodular, superficial, micronodular, morphoeaform, infiltrating.**
- **Nodular is the commonest genital subtype**: a **pearly, skin-toned papule or plaque with overlying telangiectasia** that may **ulcerate centrally** (Fig. 30.40).
- **Distant metastasis is uncommon but portends a poor prognosis.**

Treatment

- **Local excision** is primary; **Mohs micrographic surgery** should be considered to **preserve genital form and function**.
- **Metastatic, locally advanced or unresectable disease**: FDA-approved **Hedgehog signalling pathway inhibitors** — **vismodegib (2012)** and **sonidegib (2015)**.

Kaposi Sarcoma

Nature and epidemiology

- **Kaposi sarcoma (KS)** is an **angioproliferative disease of endothelial origin** that **requires human herpesvirus 8 (HHV-8)**.
- Whether it is **neoplastic or hyperplastic** is **controversial**; clonality evidence is mixed.
- **Classic KS**: a chronic disease of **elderly men of Jewish, Mediterranean or Eastern European descent**.
- **HIV-1** increased KS incidence **more than 7000-fold**:
 - It affects patients with **significant immune impairment (CD4+ T cells <500/mm³)**.
 - It affects **about 40% of homosexual men with AIDS**, versus **<5% of other risk groups**.
 - Incidence is **falling with antiretroviral therapy (ART)**.
- KS also occurs in **solid-organ transplant recipients**.

Clinical features

- **Lesions: painless, non-pruritic**; may be **mistaken for purpura or haematoma**; may **coalesce** and cause **lymphatic or venous obstruction → oedema**.
- **Glans lesions** can cause **urethral obstruction at the meatus or fossa navicularis**.
- **Genital KS is rare even with HIV** — **about 3% of patients with AIDS**.

CLASSIC VERSUS EPIDEMIC VERSUS TRANSPLANT KAPOSİ SARCOMA

Feature	Classic KS	Epidemic (AIDS-related) KS	Transplant-associated KS
Host	Elderly men ; Jewish, Mediterranean, Eastern European	HIV , CD4+ <500/mm ³ ; homosexual men with AIDS	Solid-organ recipients

Feature	Classic KS	Epidemic (AIDS-related) KS	Transplant-associated KS
Lesions	Slow blue-red to purple macules, plaques, nodules	Single lesion → disseminated cutaneous and visceral (Fig. 30.41)	Single → disseminated
Site	Lower limbs; oral and GI possible	Anywhere; a solitary genital lesion may be the first sign	Anywhere, including genitalia
Genitalia	Seldom	Possible (~3% of AIDS)	Possible
Specific treatment	Local or systemic as below	Combination ART; beware immune reconstitution inflammatory syndrome (IRIS) progression	Reduce immunosuppression or switch calcineurin inhibitor → mammalian target of rapamycin (mTOR) inhibitor — may resolve without other treatment

Treatment

- **Goals: palliative** — symptom relief, prevention of progression, **shrinkage to relieve oedema; cure may be unrealistic.**

TREATMENT OF KAPOSI SARCOMA BY EXTENT

Extent	Treatment
Solitary lesion	No local treatment necessary; if desired: excision, laser ablation, cryotherapy, imiquimod 5%, intralesional vinblastine
Extensive locoregional	Radiotherapy 15–30 Gy (high objective response)
Widely disseminated	Systemic vincristine, doxorubicin and bleomycin (treatment of choice)

Concept — KS is *HHV-8 unmasked by immunosuppression*, so the first treatment is **restoring immunity** — ART in AIDS, reduction or an mTOR switch in transplant recipients — while local and cytotoxic therapy is **palliative and scaled to extent**.

Pseudoepitheliomatous, Keratotic and Micaceous Balanitis

Features

- **Pseudoepitheliomatous, keratotic and micaceous balanitis (PEKMB)** is rare: a **thick hyperkeratotic plaque on the glans of older men**, typically men circumcised as adults; often **asymptomatic** (Fig. 30.42).
- It was thought **purely benign**, but **concurrent verrucous carcinoma** has been reported; **pre-malignant status is controversial**.
- **Histology is essential to exclude SCC and verrucous carcinoma**; it shows a **hyperplastic epidermis with ridges extending deeply into the dermis**.

Treatment

- **Local surgical excision or ablation, with close follow-up.**
- **Topical 5-fluorouracil** (anecdotal).
- **Refractory disease: glans resurfacing and split-skin graft reconstruction.**

Figure notes

- **Fig. 30.42 PEKMB:** the glans becomes covered with mica (asbestos-like) scales and horny crusts (Ferrándiz-Pulido et al., 2012).

Melanoma

Epidemiology and risk factors

- A **malignant skin cancer of melanocytes**; the **fifth most commonly diagnosed cancer in the United States**, with **rising incidence**.
- **Risk factors**: family history, genetic markers, **fair skin**, **light eye colour**, **excessive ultraviolet exposure** (**multiple blistering childhood or adolescent sunburns**).

Genital melanoma

- **Male**: uncommon — **<1% of penile cancers and of melanomas**; **urethral melanoma is rarer still**.
- **Female**: commoner — **7–10% of vulvar malignancies**, the **second commonest after SCC**; commoner in **White females**, but the **prognosis is worse in Black females**.
- **Presentation**: a **black, blue or brown macule or papule with an irregular border**; may be **unpigmented or ulcerated**.
- **Early diagnosis is critical**: superficial lesions treated by **wide local excision or partial penectomy** have **excellent control**; the same applies in females.
- **Metastatic disease** was historically **universally poor** despite surgery and multi-agent chemotherapy; **targeted and immune therapy have transformed advanced disease**.

SYSTEMIC THERAPY FOR ADVANCED MELANOMA

Agent	Class / target
Vemurafenib, dabrafenib	BRAF inhibitors
Trametinib	MAP-ERK kinase (MEK) inhibitor
Ipilimumab	Immune checkpoint inhibitor (cytotoxic T-lymphocyte-associated protein 4, CTLA-4)
Nivolumab, pembrolizumab (formerly lambrolizumab)	Anti-programmed cell death protein 1 (PD-1) immune checkpoint inhibitors

Extramammary Paget Disease

Epidemiology

- **Extramammary Paget disease (EPD)** is **uncommon**, arising at **apocrine gland-bearing sites**, in the **sixth to eighth decades**, with a **female predominance**; the **penis and scrotum** can be involved.
- **Commonest site**: the **vulva in females**; the **perianal region** is also affected in males.

Associated malignancy

- An **underlying malignancy** is present in at least **10–30%**.
- In **males**: associated with **urethral, prostate, bladder, rectal and apocrine carcinoma**.
- A **systematic search for underlying carcinoma** is **mandatory in every case**.

Clinical features and diagnosis

- **Lesion**: an **erythematous plaque with a sharp border** (Fig. 30.43); may be **asymptomatic, pruritic or burning**.
- **Histology**: **vacuolated Paget cells in the epidermis ("buckshot scatter")**, staining for **glandular cytokeratins**, **epithelial membrane antigen (EMA)** and **carcinoembryonic antigen (CEA)**.

Treatment

- **Surgical excision or Mohs micrographic surgery**.
- **Alternatives**: **radiotherapy, photodynamic therapy, topical imiquimod 5% or 5-fluorouracil**.

Cutaneous T-Cell Lymphoma

Definition and scope

- **Cutaneous T-cell lymphomas (CTCLs)** are a heterogeneous group of T-cell neoplasms presenting in the skin without extracutaneous disease at diagnosis.
- **Members:** mycosis fungoides, Sézary syndrome, lymphomatoid papulosis, pagetoid reticulosis.
- **Risk is increased with HIV.**
- **Genital involvement** occurs in both sexes, usually with extragenital disease.
- **CTCL is the majority of primary cutaneous lymphomas;** B-cell lymphomas account for only 20–25%.
- **Biopsy histopathology is definitive.**

Clinical course

- **Initial pruritic patches** must be distinguished from psoriasis, eczema, superficial fungal infection and drug reaction.
- **Early lesions** favour the suprapubic and buttock skin in both sexes (Fig. 30.44).
- **Progression:** patches → plaques → erosions and ulcers → tumours; haematological involvement = Sézary syndrome; chronic, over many years.

TREATMENT OF CUTANEOUS T-CELL LYMPHOMA

Category	Options
Topical	Ultrapotent corticosteroids, nitrogen mustard, carmustine (a majority achieve complete remission)
Radiotherapy	Including total skin electron beam
Phototherapy	PUVA, narrow-band UVB
Systemic	Chemotherapy, interferons, retinoids

Figure notes

- **Fig. 30.44** Mycosis fungoides of the buttocks: (A) **limited plaque stage;** (B) **advanced — plaques, patches and tumours** (Bolognia et al., 2003).

TOPICAL AND TISSUE-SPARING THERAPY ACROSS GENITAL NEOPLASIA

Entity	Mohs	Topical 5-FU	Imiquimod 5%	Other topical / local
SCCis	Yes	Yes	Yes	Laser, cryoablation
Bowenoid papulosis	—	0.5%	Yes	Tazarotene 0.5%; ablation
Verrucous carcinoma	Yes	—	—	No primary radiotherapy
BCC	Yes	—	—	Hedgehog inhibitors if advanced
KS (solitary)	—	—	Yes	Intralesional vinblastine; laser, cryo
PEKMB	—	Yes (anecdotal)	—	Excision, ablation; glans resurfacing
EPD	Yes	Yes	Yes	Radiotherapy, photodynamic therapy
CTCL	—	—	—	Ultrapotent steroid, nitrogen mustard, carmustine

- **Pattern:** genital neoplasia repeatedly favours **tissue-sparing** options (Mohs, topical field therapy); the trade-off is **higher recurrence than excision**.

Concept — Genital neoplasia **balances oncological clearance against preservation of genital form and function — hence Mohs and topical field therapy throughout — and biopsy of several areas is mandatory because the in situ, verrucous and inflammatory mimics look alike on the surface.**

BENIGN CUTANEOUS DISORDERS SPECIFIC TO THE MALE GENITALIA

Angiokeratoma of Fordyce

- **Benign vascular ectasias of dermal vessels on the penis and scrotum of adults.**
- **Lesion:** multiple 1–2 mm red or purple papules (Fig. 30.45A), sometimes with **generalised scrotal redness**.
- Usually asymptomatic, but **bleeding can occur**.
- **Similar lesions occur in Fabry disease, a lysosomal storage disorder** (Fig. 30.45B).
- **Treatment** is usually unnecessary; **laser photocoagulation** (erbium:YAG, neodymium:YAG, potassium titanyl phosphate (KTP), argon) has been used in selected cases.

Pearly Penile Papules

- **Lesion:** white, dome-shaped or filiform, closely spaced papules on the coronal ring of the glans (Fig. 30.45C).
- **Prevalence:** common — 14–48% of young postpubertal men, especially the **uncircumcised**.
- **Often misdiagnosed as condyloma**, but there is **no evidence of an HPV role** and **no association with cervical intraepithelial neoplasia in partners**.
- **Histology:** angiofibromas, similar to the **facial angiofibromas of tuberous sclerosis**.
- **Management:** reassure — a benign normal variant; for cosmetic concern, **CO₂ laser or cryotherapy**.

Zoon Balanitis

- **Synonyms:** plasma cell balanitis, balanitis plasmacellularis.
- **Demography:** middle-aged or elderly **uncircumcised men**; cause **unknown**.
- **Lesion:** smooth, shiny, moist, erythematous, well-circumscribed papules and plaques on the glans (Fig. 30.45D); shallow erosions may occur; lesions reach **up to 2 cm**.
- **Symptoms:** asymptomatic, or **mild pruritus or tenderness**.
- **SCC and EPD must be excluded, typically by biopsy**.
- **Treatment:**
 - **Circumcision prevents the disease and cures most cases.**
 - For men averse to circumcision: **topical corticosteroids** (symptomatic), **topical calcineurin inhibitors** (tacrolimus, pimecrolimus), **topical retinoids, laser**.

ZOON BALANITIS VERSUS ERYTHROPLASIA OF QUEYRAT

Feature	Zoon balanitis	Erythroplasia of Queyrat
Host	Middle-aged or elderly uncircumcised	Uncircumcised , older
Lesion	Shiny, moist, well-circumscribed red glans plaques	Red glans plaque
Nature	Benign, inflammatory (plasma cell)	SCCis (HPV 8/16; LS, LP)
Key step	Biopsy to exclude SCC and EPD	Biopsy, several areas, to exclude invasion
Cure	Circumcision	Excision / Mohs / ablation ; topical 5-FU or imiquimod

Sclerosing Lymphangitis

- **Nonvenereal sclerosing lymphangitis** is a **rare, indurated, firm, slightly tender cord-like lesion** of the **coronal sulcus** and adjacent skin; usually **flesh-coloured**, occasionally red.
- **Mechanism:** proposed **lymphatic thrombosis**.
- **Demography:** age 20–40, **sexually active**; associated with **vigorous sexual activity**.

- **Resolves within several weeks.**
- **Consider screening for gonococcal and non-gonococcal urethritis** (somewhat controversial).

Median Raphe Cysts

- **Young men; on the ventral penis, most often near the glans.**
- Arise from **aberrant urethral epithelium** but **do not communicate with the urethra.**
- **Treatment: surgical excision under local anaesthesia.**

Ectopic Sebaceous Glands

- **Pin-sized, flesh-coloured papules on the penile shaft** (Fig. 30.45E), mistaken for verruca.
- **No treatment; reassurance suffices.**

BENIGN MALE GENITAL LESIONS AT A GLANCE

Lesion	Site	Appearance	Mimics	Management
Angiokeratoma of Fordyce	Scrotum, penis	1–2 mm red or purple papules, multiple	Fabry disease	Observe; laser if bleeding or cosmetic
Pearly penile papules	Coronal ring	White dome or filiform papules in rows	Condyloma	Reassure; CO ₂ laser or cryo if desired
Ectopic sebaceous glands	Penile shaft	Pin-sized flesh-coloured papules	Verruca	Reassure
Median raphe cyst	Ventral penis near glans	Cyst	Urethral diverticulum (non-communicating)	Excision under local anaesthesia
Sclerosing lymphangitis	Coronal sulcus	Firm cord	—	Self-limiting (weeks); screen for urethritis
Zoon balanitis	Glans	Shiny moist red plaque	SCCs, EPD	Biopsy → circumcision

Concept — Most of these are **normal variants recognised by site and morphology**; the skill lies in **not treating them as condyloma or cancer** — reassurance is the treatment — with **Zoon balanitis the exception**, because its red glans plaque obliges biopsy to exclude SCCs.

KEY POINTS

- **Diagnosis rests on history and examination; extragenital findings may be the key**, so perform a **full skin survey**.
- **Topical corticosteroid adverse effects are significant**, both systemic and local, and **worse under the foreskin (occlusion)**; on genital skin, **use low-potency agents for short courses** in general.
- **Genital dermatoses fall into categories: allergic, papulosquamous, vesiculobullous, ulcerative, infectious, neoplastic and miscellaneous.**
- **Biopsy separates look-alike diseases and excludes malignancy.**
- **Local modalities (laser, photodynamic therapy, ultraviolet, cryotherapy) are alternatives to excision** in some cases.